



Beef Northwest - Project overview



Date: 05/29/2024

Project objective:

To develop an AI-driven chatbot designed to optimize cattle feed rationing and provide predictive insights on cattle treatment, aimed at promoting early output delivery and improved wellness. The chatbot will feature advanced visualization tools, including graphs and charts, to facilitate easy understanding and decision-making. The primary users will be the company's staff, specifically on-field cowboys and veterinary doctors.

Project scope:

1. Chatbot Development:

- Design and implement an AI chatbot capable of handling queries related to cattle feed optimization and cattle treatment predictions.
- Ensure the chatbot is intuitive and user-friendly for on-field cowboys and veterinary doctors.

2. Cattle Feed Optimization:

- Integrate algorithms to analyze and optimize cattle feed rationing.
- Provide actionable insights to enhance early output delivery of cattle.
- Primary goal is to guide decisions that result in cattle reaching their target weight more quickly

3. Cattle Treatment Predictions:

- Develop predictive models to offer treatment recommendations for cattle health issues.
- Focus on early wellness and preventive care.
- Primary goal is to measurably reduce mortality, and make better decisions more quickly regarding cattle that are not going to survive

4. Visualization Tools:

- Implement features to display data through interactive graphs and charts.
- Ensure visualizations are clear and aid in decision-making.

5. Data Integration and Management:

- Integrate with existing data sources such as Snowflake and MySQL databases.
- Ensure seamless data import, processing, and usage within the chatbot framework.

6. User Access and Roles:

- Restrict access to the company's staff, specifically tailored for use by on-field cowboys and veterinary doctors.

7. Testing and Validation:

- Conduct thorough testing of the chatbot for functionality, accuracy, and user experience.
- Validate the chatbot's performance in real-world scenarios with actual end-users.

8. POC Delivery:

- Offer video of working POC chatbot to the client.

Key features:

1. Basic chatbot functionality
2. Visualization- Charts and graphs
3. Restricted access
4. Download charts
5. Copy response
6. Record user-experience response
7. Maintain chat history

Challenges:

1. Model selection

The team has commenced development using the OpenAI model. However, the client has expressed concerns about data security and is evaluating the choice between OpenAI and Gemini models. Should the client prefer Gemini, the team will need to undertake significant rework, including data preparation and API integration adjustments. This may impact the project timeline.

2. Unsorted Data

The data provided by the client is currently unsorted, posing a challenge for the team. Identifying and organizing the various data sets will require additional effort and meticulous planning. Appropriate measures will be developed to manage and process the unsorted data effectively, ensuring its reliability and usability for model training and predictions.

Sample queries:

Cattle Feed Queries:

1. What is the optimal feed ration for young calves.
2. How can I adjust the feed to improve weight gain of Lot X.
3. What feed ingredients are best for high protein content.
4. Can you provide a feeding schedule for lactating cows.
5. How do I reduce feed costs while maintaining quality.
6. What are the nutritional requirements for beef cattle in winter.
7. Can you compare the feed efficiency of different feed types.
8. How much feed should I give to weaning calves.
9. What supplements are necessary for cows.
10. Can you show a graph of feed consumption trends over the past month.

Cattle Treatment Queries:

1. Cattle is facing [Symptom], which medicine should be given?
2. How much dose of [Medicine] should be given for [disease].
3. How do I treat respiratory infections in cattle.
4. Which vaccinations are due for next month.
5. Can you provide a treatment plan for mastitis.
6. What are the best practices for preventing hoof rot.
7. What amount of daily dose of [medicine] should be provided to cattle suffering from [disease].
8. What are the signs of nutritional deficiencies in cattle.
9. What are the recent health issues in the cattle.

10. What are the common treatments for parasitic infections.
11. How often should I deworm my cattle.

